

Chapter 1

Fields and Vector Spaces

1.1 Fields

Definition 1.1 (Field). Let F be a set with two binary operations, addition $+: F \times F \rightarrow F$ and multiplication $\cdot: F \times F \rightarrow F$. We say that F is a *field* if it satisfies the following axioms for all $a, b, c \in F$:

Axioms for Addition $(F, +)$:

A1. Associativity: $(a + b) + c = a + (b + c)$

A2. Commutativity: $a + b = b + a$

A3. Additive Identity: There exists an element $0 \in F$ such that $a + 0 = a$.

A4. Additive Inverse: For each $a \in F$, there exists an element $-a \in F$ such that $a + (-a) = 0$.

Axioms for Multiplication $(F, \cdot)$:

M1. Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

M2. Commutativity: $a \cdot b = b \cdot a$

M3. Multiplicative Identity: There exists an element $1 \in F$, with $1 \neq 0$, such that $a \cdot 1 = a$.

M4. Multiplicative Inverse: For each $a \in F \setminus \{0\}$ there exists an element $a^{-1} \in F \setminus \{0\}$ such that $a \cdot a^{-1} = 1$.

Distributivity:

D1. $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

Remark. This is all the structure we need from the real numbers to do linear algebra. We can do linear algebra with many other fields, not just $\mathbb{R}$, like the finite field $\mathbb{F}_2$. Abstracting the scalars into an arbitrary field F allows for some rich and interesting results in linear algebra.

Example 1.2. The integers modulo p , $\mathbb{F}_p := \{0, 1, \dots, p-1\}$, where p is prime, form a field. Addition and multiplication are defined modulo p . All properties except the existence of a multiplicative inverse follow easily from the fact that $\mathbb{Z}$ is a commutative ring.

Lemma 1.3. In $\mathbb{F}_p$, $xy = 0 \iff x = 0$ or $y = 0$.

Proof. $xy = 0 \implies p \mid xy \implies p \mid x$ or $p \mid y$ (by Euclid's lemma). This means $x \equiv 0 \pmod{p}$ or $y \equiv 0 \pmod{p}$. $\square$

Theorem 1.4. For every $a \in \mathbb{F}_p \setminus \{0\}$ there exists $a^{-1} \in \mathbb{F}_p \setminus \{0\}$ such that $aa^{-1} = 1 = a^{-1}a$.

Proof. Given $a \in \mathbb{F}_p$, by commutativity we only need to find a $b \in \mathbb{F}_p$ such that $ab = 1$. Finding an explicit construction for b is difficult, so we instead show that the set of multiples $S := \{a \cdot 0, a \cdot 1, \dots, a \cdot (p-1)\}$ is exactly equal to $\mathbb{F}_p$.

It suffices to show that the p elements $a \cdot 0, a \cdot 1, \dots, a \cdot (p-1) \in \mathbb{F}_p$ are all distinct. Suppose $a \cdot c = a \cdot d$. Then $a \cdot (c - d) = 0$.

By our lemma, since $a \neq 0$ ($p \nmid a$), we must have $c - d = 0$, which implies $c = d$. Thus, multiplication by a maps distinct elements to distinct elements. Since there are p elements in $\mathbb{F}_p$, the set S covers all of $\mathbb{F}_p$. Because $1 \in \mathbb{F}_p$, there must be some element $b \in \{0, \dots, p-1\}$ such that $ab = 1$. $\square$

Corollary 1.5. $\mathbb{F}_p$ is a field.

Remark. The proof above relies on the Pigeonhole Principle. Because multiplication by a non-zero element a is injective on the finite set $\mathbb{F}_p$, it must also be surjective.

Definition 1.6 (Subfield). A subset $F \subseteq E$ of a field E is a *subfield* if $1 \in F$ and it is closed under arithmetic: whenever $a, b \in F$ we have $a + b, ab \in F$; if $a \in F$ then $-a \in F$; and if $a \in F \setminus \{0\}$ then $a^{-1} \in F$.

Remark. This definition is slightly redundant. To prove a subset $F \subseteq E$ is a subfield, it actually suffices to show closure under addition and multiplication, the existence of multiplicative inverses for non-zero elements, and that $-1 \in F$. Because if $-1 \in F$ and we have the closures above, then $(-1)(-1) = 1 \in F$ and $(-1) + 1 = 0 \in F$. Additive inverses follow from closure of addition with -1 .

Example 1.7. Some examples of fields and subfields are:

- $\mathbb{Q}[i] := \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$ is a field (the "Gaussian rationals"). It is a subfield of $\mathbb{C}$.
- $\mathbb{Q}[\sqrt{2}] := \{a + b\sqrt{2} : a, b \in \mathbb{Q}\} \subseteq \mathbb{R}$ is a field, and it is a subfield of $\mathbb{R}$ (exercise: show this).
- The field of rational functions $F(X) := \left\{ \frac{p(X)}{q(X)} : p, q \in F[X], q \neq 0 \right\}$ is a field. $\mathbb{Q}(x)$ is a subfield of $\mathbb{R}(x)$, for example.

1.2 Vector Spaces

Definition 1.8. The vector space F^n of n -tuples over a field F is defined as:

$$F^n := \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} : x_1, \dots, x_n \in F \right\}$$

Remark. This is the primordial example of a vector space. Every finite-dimensional vector space is structurally identical (isomorphic) to F^n . The geometric intuition of $\mathbb{R}^n$ (arrows, scaling, and adding tip-to-tail) carries over algebraically to arbitrary fields.

Definition 1.9 (Commutative Group). A commutative group $(V, +)$ is a set equipped with a binary operation $+: V \times V \rightarrow V$ satisfying:

- **Associativity:** For all $u, v, w \in V$, $u + (v + w) = (u + v) + w$.
- **Commutativity:** For all $v, w \in V$, $v + w = w + v$.
- **Additive Identity:** There exists $0 := (0, \dots, 0) \in V$ such that for all $v \in V$, $v + 0 = 0 + v = v$.
- **Additive Inverse:** For all $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0 = (-v) + v$.

Lemma 1.10. In a commutative group $(V, +)$, the zero vector and additive inverses are unique.

Proof. Given another zero vector $0'$, we have $0' = 0 + 0' = 0$. Thus, the zero vector is unique.

If w also acts as an additive inverse to v (so $w + v = 0$), then:

$$w = w + 0 = w + (v + (-v)) = (w + v) + (-v) = 0 + (-v) = -v.$$

Thus, the additive inverse is unique. □

Definition 1.11 (Vector Space). A vector space V over a field F is a commutative group $(V, +)$ together with an action $F \times V \rightarrow V$, mapped by $(\lambda, v) \mapsto \lambda v$, called *scalar multiplication*, satisfying the following properties for all $\lambda, \mu \in F$ and $v, w \in V$:

- **Associativity:** $\lambda(\mu v) = (\lambda\mu)v$
- **Distributivity of scalar sums:** $(\lambda + \mu)v = \lambda v + \mu v$
- **Distributivity of vector sums:** $\lambda(v + w) = \lambda v + \lambda w$
- **Unit multiplication:** $1 \cdot v = v$

Remark. Structurally, one can think of a vector space as having two tiers: the “ground” field F provides the scalars (stretching factors), while the commutative group V provides the vectors (the objects being stretched and added; indeed, F acts on V to get scalar multiples of vectors). The axioms ensure these tiers interact in a linear way.

Example 1.12. Other examples of vector spaces include:

- $F[X] = \{a_0 + a_1X + \cdots + a_nX^n : a_i \in F\}$ (polynomials).
- Subfields $F \subseteq E$ (E acts as a vector space over its subfield F).
- $\text{Fun}(X, F) = \{f : X \rightarrow F\}$ (functions from an arbitrary set X into F).